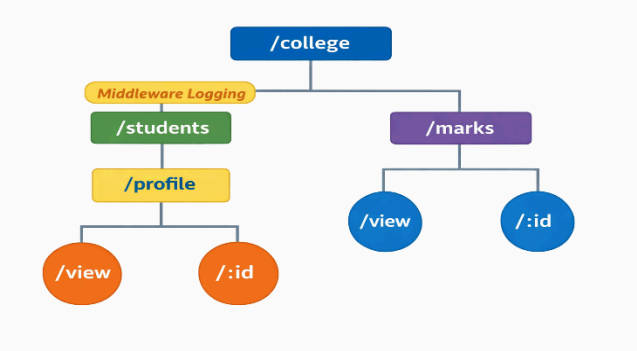
Experiment 4: Hierarchical Routing in Express.js

Aim:
To design and configure hierarchical routing in Express.js using a modular approach with separate routes, enabling structured URL handling and middleware application.



Procedure

1. Create project
mkdir hierarchical-routing
cd hierarchical-routing

2. Initialize project
npm init -y

3. Install Express
npm install express

4. Create files

routes/college.js
routes/marks.js

routes/profile.js

routes/students.js
app.js

college-app/

│

├── app.js

└── routes/

├── college.js

├── students.js

├── profile.js

└── marks.js

App.js:

```
const express = require('express');

const app = express();

const collegeRouter = require('./routes/college');

app.use('/college', collegeRouter);

app.listen(3000, () => {
  console.log('Server running on port 3000');
});
```

routes/college.js

```
const express = require('express');

const router = express.Router();

const studentsRouter = require('./students');
const marksRouter = require('./marks');

// Middleware at /college level
router.use((req, res, next) => {
  console.log('College middleware:', req.method, req.originalUrl);
  next();
});

router.use('/students', studentsRouter);
router.use('/marks', marksRouter);

module.exports = router;
```

routes/students.js

```
const express = require('express');

const router = express.Router();
const profileRouter = require('./profile');

router.use('/profile', profileRouter);

module.exports = router;
```

routes/profile.js

```
const express = require('express');

const router = express.Router();

// Middleware only for profile routes
router.use((req, res, next) => {
  console.log('Profile middleware');
  next();
});

router.get('/view', (req, res) => {
  res.send('Student Profile View');
});

router.get('/:id', (req, res) => {
  res.send(`Student Profile ID: ${req.params.id}`);
});

module.exports = router;
```

routes/marks.js

```
const express = require('express');

const router = express.Router();

router.get('/view', (req, res) => {
  res.send('Marks View');
});

router.get('/:id', (req, res) => {
  res.send(`Marks for Student ID: ${req.params.id}`);
});

module.exports = router;
```

OUTPUT:

URL	Handler
/college/students/profile/view	Student profile view
/college/students/profile/101	Profile of student 101
/college/marks/view	Marks view
/college/marks/101	Marks of student 101